.

Curriculum Vitae

Tomás Méndez Echenagucia Ph.D.
Associate Professor
February 2026

University of Washington
Department of Architecture
College of Built Environments
tmendeze@uw.edu
https://tmsmendez.github.io/

3950 University Way NE
98105 Seattle, WA

.

I Career Summary

I.A Education

2011 - 2014	**Doctor of Philosophy** Politecnico di Torino - Italy Architecture and Building Design (March 2014) Dissertation: “Computational Search in Architectural Design” Advisors: Prof. Mario Sassone, Prof. Arianna Astolfi and Prof. Pierre-Alain Croset Awarded the title of “Excellence” by examining board
2004 - 2007	**Master’s degree** (Dual degree) Politecnico di Torino - Italy Architecture and Construction Thesis: “Forma acustica, architettonica e strutturale” Advisors: Arch. Maarten Jansen, Prof. Mario Sassone and Prof. Arianna Astolfi
2000 - 2007	**Architecture Degree** (Dual degree) Universidad Central de Venezuela - Venezuela Architecture Thesis: “Forma acustica, architettonica e strutturale” Advisors: Arch. Maarten Jansen, Prof. Mario Sassone and Prof. Arianna Astolfi
2000	**Summer design studio** Columbia University, United States of America Introduction to Architecture summer course in New York City Design of a fine arts museum in the meat packing district Advisor: Prof. Mojdeh Baratloo

I.B Academic and Professional Experience

2024 - present	**Associate Professor** University of Washington Department of Architecture College of Built Environments
2023 - present	**Director - Master of Science Program in Architecture (Design Technology)** University of Washington Department of Architecture College of Built Environments
2023 - present	**Co-Director - UW Center for Digital Fabrication** University of Washington

2019 - 2024	**Assistant Professor** University of Washington Department of Architecture College of Built Environments
2014 - 2019	**Post-doctoral researcher** ETH Zürich Block Research Group (BRG) Institute of Technology in Architecture
2011 - 2014	**PhD Student / Teaching Assistant** Politecnico di Torino Department of Architecture and Design
2008 - 2014	**Co-Founder / Lead Architect** TMS Arquitectura Caracas - Venezuela & Turin - Italy
2007 - 2008	**Architect / Project manager** Arquitectura Multimedia Caracas - Venezuela
2005 - 2007	**Architectural Intern** Frlan+Jansen Architetti Turin - Italy

I.C Awards and Recognitions

2022	**AIA Seattle Honors** Award of Merit With Vidhya Rajendran (UW) and Ryan Mullenix (NBBJ) In the category "Research & Innovation"
2022	**ARC Award 2022** - Swiss Architecture Award Block Research Group - ETH Zürich Awarded to the HiLo Research Unit in the category "Digitalization."
2022	**IABSE Projects and Technology Awards (shortlisted)** International Association for Bridge and Structural Engineering Block Research Group - ETH Zürich
2021	**Fast Company's Most Creative People** Selected for work on acoustic meta-materials with NBBJ and Vidhya Rajendran

2017	**The Structural Awards 2017 (shortlisted)** The Institution of Structural Engineers - UK Block Research Group - ETH Zürich The Armadillo Vault was shortlisted in the categories of structural artistry and small projects, receiving a commendation for the latter.
2016	**DETAIL Prize 2016** *DETAIL* magazine - DE Block Research Group - ETH Zürich Readers' Prize and acknowledgment in the category "Structures"
2008	**Hangai Prize - IASS** Awarded by the International Association for Spatial and Shell Structures (IASS) to young researchers for innovative papers Awarded paper: **Méndez Echenagucia, T.** et al. (2008)

I.D Professional Licensure and Certification

2011 - 2019	**Ordine degli Architetti di Torino** Turin - Italy
2007 - present	**Colegio de Ingenieros de Venezuela*** Caracas - Venezuela licensed as an Architect

*Venezuelan architecture certification is awarded by the guild of engineers (Colegio de Ingenieros) to those who have obtained a five year degree in architecture from a certified architecture school without examination.

II Research

II.A Publications

Peer-reviewed journals

Méndez Echenagucia, T., Astolfi, A., Jansen, M., and Sassone, M. (2008). "Architectural acoustic and structural form." *Journal of the International Association for Shell and Spatial Structures*, 49(159):181–186. ISSN 1028365X.

Palma, M., Sarotto, M., **Méndez Echenagucia, T.**, Sassone, M., and Astolfi, A. (2014). "Sound strength driven parametric design of an acoustic shell in a free field environment." *Building Acoustics*, 21(1):31–42. ISSN 1351010X.

Méndez Echenagucia, T., Sassone, M., Astolfi, A., Shtrepi, L., and Van der Harten, A. (2014). "EDT , C 80 and G driven auditorium design." *Building Acoustics*, 21(1):43–54.

Méndez Echenagucia, T., Capozzoli, A., Cascone, Y., and Sassone, M. (2015). "The early design stage of a building envelope: Multi-objective search through heating, cooling and lighting energy performance analysis." *Applied Energy*, 154:577–591. ISSN 03062619.

Block, P., Schlueter, A., Veenendaal, D., Bakker, J., Begle, M., Hischier, I., Hofer, J., Jayathissa, P., Maxwell, I., **Méndez Echenagucia, T.** , Nagy, Z., Pigram, D., Svetozarevic, B., Torsing, R., Verbeek, J., Willmann, A., and Lydon, G. (2017). "NEST HiLo: Investigating lightweight construction and adaptive energy systems." *Journal of Building Engineering*, 12:332–341.

Roozen, N., Leclère, Q., Urbán, D., **Méndez Echenagucia, T.**, Block, P., Rychtárikovà, M., and Glorieux, C. (2018). "Assessment of the airborne sound insulation from mobility vibration measurements; a hybrid experimental numerical approach." *Journal of Sound and Vibration*, 432:680–698.

Méndez Echenagucia, T., Pigram, D., Liew, A., Mele, T.V., and Block, P. (2019). "A cable-net and fabric formwork system for the construction of concrete shells: Design, fabrication and construction of a full scale prototype." *Structures*, 18:72 – 82. ISSN 2352-0124. Advanced Manufacturing and Materials for Innovative Structural Design.

Shtrepi, L., **Méndez Echenagucia, T.**, Badino, E., and Astolfi, A. (2020). "A performance-based optimization approach for diffusive surface topology design." *Building Acoustics*.

López López, D., Roca, P., Liew, A., **Méndez Echenagucia, T.**, Van Mele, T., and Block, P. (2021). "A three-dimensional approach to the extended limit analysis of reinforced masonry." *Structures*. ISSN 2352-0124.

Rajendran, V., Piacsek, A., and **Méndez Echenagucia, T.** (2022). "Design of broadband helmholtz resonator arrays using the radiation impedance method." *The Journal of the Acoustical Society of America*, 151(1):457–466.

Méndez Echenagucia, T., Moroseos, T., and Meek, C. (2023). "On the tradeoffs between embodied and operational carbon in building envelope design: The impact of local climates and energy grids." *Energy and Buildings*, 278:112589. ISSN 0378-7788.

Kuehne, L.M., Habtour, E., **Méndez Echenagucia, T.**, and Orfield, S.J. (2024). "Technology-forcing to reduce environmental noise pollution: a prospectus." *Journal Of Exposure Science And Environmental Epidemiology*.

Moroseos, T., **Méndez Echenagucia, T.** , and Meek, C. (2025). "Impact of a dynamic grid mix and climate on operational carbon emissions modeling for different building typologies and climate zones." *Energy and Buildings*, 342:115854. ISSN 0378-7788.

Huang, M., Ellingboe, E., Lin, M.Y., **Méndez Echenagucia, T.**, and Simonen, K. (2026). "Tool requirements for life cycle assessment in the innovation of novel carbon-storing construction materials." *Applied Sciences*, 16(8). ISSN 2076-3417.

Iyer, H., Tabatabaei Manesh, M., Dillon, M., Kim, R., Kim, D., and **Méndez Echenagucia, T.** Roumeli, E. (2026). "Biodegradable, self-extinguishing foams made from ulva for broadband acoustic absorption." *Advanced Materials*. **(In Review)**.

Nanayakkara, K. and **Méndez Echenagucia, T.** (2026). "Analysis of timber connections using discontinuities and stress fields." *Special issue "Advances in Structural Optimisation with Practical Applications " of Engineering and Computational Mechanics*. **(Invited paper - abstract in review)**.

Tabatabaei Manesh, M. and **Méndez Echenagucia, T.** (2026). "A surrogate modeling framework for the design of fano-like ventilated acoustic barriers." *Applied Acoustics*. **(In Review)**.

Méndez Echenagucia, T., Piacsek, A., and Roozen, N. (2026a). "Laser doppler vibrometry and fea studies of the sound transmission of cross laminated timber samples with varying geometries." *Journal of Sound and Vibration*. **(In Preparation)**.

Méndez Echenagucia, T., Tabatabaei Manesh, M., and Kam-Ming, M.T. (2026b). "Convolutional and graph based neural network models for the vibro-acoustic behaviour of timber floor slabs." *Journal of Sound and Vibration*. **(In Preparation)**.

Peer-reviewed conferences with proceedings

Sassone, M., **Méndez Echenagucia, T.**, and Pugnale, A. (2008). "On the interaction between architecture and engineering: the acoustic optimization of a reinforced concrete shell." In J. Abel and R. Cooke, editors, *Proceedings of the 6th International Conference on Computation of Shell and Spatial Structures IASS-IACM 2008: "Spanning Nano to Mega"*. May 28 - 31.

Méndez Echenagucia, T., Astolfi, A., Jansen, M., and Sassone, M. (2008). "Architectural, acoustic and structural form." In *The IASS-SLTE 2008 Symposium "Shell and Spatial Structures: New Materials and Technologies, New Designs and Innovations – A Sustainable Approach to Architectural and Structural Design"*. October 27 - 31.

Méndez Echenagucia, T., Astolfi, A., Sassone, M., Shtrepi, L., and Van der Harten, A. (2012a). "Esplorazione Multi obiettivo nella progettazione acustica architettonica." In *Convegno nazionale AIA 2012*, pages 1–6. Rome. July 4 - 6.

Méndez Echenagucia, T., Pugnale, A., and Sassone, M. (2012b). "Multi-Objective Optimization of Concrete Shells." In P. Cruz, editor, *Structures and Architecture. Concepts, Application and Challenges*, pages 217–218. CRC Press/Balkema, Leiden. July 24 - 26.

Méndez Echenagucia, T. and Sassone, M. (2012). "Multi-Objective Oriented Design of Shell Structures." In *IASS-APCS Symposium 2012: From Spatial Structures to Space Structures*, pages 1–7. May 21 - 24.

Méndez Echenagucia, T., Sassone, M., Astolfi, A., and Croset, P.a. (2012c). "Multi-Objective Search in the Early Phase of Architectural Design." In P. Cruz, editor, *Structures and Architecture. Concepts, Application and Challenges*, pages 341–342. CRC Press/Balkema, Leiden. July 24 - 26.

Palma, M., Sarotto, M., **Méndez Echenagucia, T.**, Sassone, M., and Astolfi, A. (2013). "Sound strength driven parametric design of an acoustic shell in a free field environment." In *International Symposium on Room Acoustics 2013*, pages 1–10. ISSN 1351010X. June 9-11.

Méndez Echenagucia, T., Astolfi, A., Sassone, M., Shtrepi, L., and Harten, A.V.D. (2013a). "NURBS and Mesh geometry in Room Acoustic Ray-tracing Simulation." In *AIA-DAGA 2013 Conference on Acoustics*, pages 1–4. Merano. March 18-21.

Méndez Echenagucia, T., Sassone, M., Astolfi, A., Shtrepi, L., and Van der Harten, A. (2013b). "Interactive Design Methods for Complex Curved Reflectors in Concert Halls." In *International Symposium on Room Acoustics 2013*, pages 1–10. June 9-11.

Méndez Echenagucia, T., Sassone, M., Astolfi, A., Shtrepi, L., and Van der Harten, A. (2013c). "EDT , C 80 and G driven auditorium design." In *International Symposium on*

Room Acoustics 2013, pages 1–10. June 9-11.

Rajabzadeh, S., Sassone, M., **Méndez Echenagucia, T.**, Rosada, A., and Rian, I. (2014). "The BRICKSHELL Meditation Centre : A Collaborative Masonry Project." In *Proceedings of the IASS-SLTE 2014 Symposium "Shells, Membranes and Spatial Structures: Footprints"*, pages 1–10. September 15 - 19.

Méndez Echenagucia, T., Sassone, M., Astolfi, A., Shtrepi, L., and Van der Harten, A. (2014). "Multi-Objective Acoustic and Structural design of shell structures for concert halls." In *Proceedings of the IASS-SLTE 2014 Symposium "Shells, Membranes and Spatial Structures: Footprints"*, pages 1–12. September 15 - 19.

Méndez Echenagucia, T. and Block, P. (2015). "Acoustic optimization of funicular shells." In *IASS 2015 Amsterdam Symposium: Future Visions*, pages 1–13. August 17 - 20.

Rippmann, M., Van Mele, T., Popescu, M., Augustynowicz, E., **Méndez Echenagucia, T.**, Calvo Barentin, C., Frick, U., and Block, P. (2016). "The Armadillo Vault: Computational design and digital fabrication of a freeform stone shell." In *Advances in Architectural Geometry 2016*, pages 344–363. September 12 - 13.

Méndez Echenagucia, T. and Roozen, B. and Block, P. (2016). "Minimization of sound radiation in doubly curved shell structures by means of stiffness." In *Proceedings of the International Association for Shell and Spatial Structures (IASS) Symposium 2016*. Tokyo, Japan. September 26 - 30.

Van Mele, T., Mehrotra, A., **Méndez Echenagucia, T.**, Frick, U., Augustynowicz, E., Ochsendorf, J., DeJong, M., and Block, P. (2016). "Form finding and structural analysis of a freeform stone vault." In T.T. K. Kawaguchi M. Ohsaki, editor, *Proceedings of the International Association for Shell and Spatial Structures (IASS) Symposium 2016*. Tokyo, Japan. September 26 - 30.

Méndez Echenagucia, T., Pigram, D., Liew, A., Van Mele, T., and Block, P. (2018). "Full-scale prototype of a cable-net and fabric formed concrete thin-shell roof." In *Proceedings of the IASS Symposium 2018*. Boston. July 16-20.

Shtrepi, L., **Méndez Echenagucia, T.**, Badino, E., and Astolfi, A. (2019). "Optimizing diffusive surface topology through a performance-based design approach." In *Proceedings of the International Symposium on Room Acoustics*. Amsterdam. September 15-17.

Bouvet, G., Shtrepi, L., Bo, E., **Méndez Echenagucia, T.**, and Astolfi, A. (2020). "Computational Design: Acoustic shells for Ancient Theatres." In *Proceedings of the FA2020 Conference*. Lyon. December 7 - 11.

Masmeijer, T., **Méndez Echenagucia, T.**, Slavič, J., Loendersloot, R., and Habtour, E.

(2023). “Effect of eccentricity on sensing in spider web inspired cable nets.” In *EURODYN 2023 - XII International Conference on Structural Dynamics*. Delf - The Netherlands. July 2 - 5.

Rajendran, V., **Méndez Echenagucia, T.**, and Piacsek, A. (2023). “Design of sound absorptive metamaterials with shared waveguides by means of numerical analysis and analytical modeling.” In *Forum Acusticum 2023 - Acoustics for a green world*. Turin - Italy. September 11 - 15.

Recart, C., **Méndez Echenagucia, T.**, and Sturts Dossick, C. (2023). “Impacts of air cavities on hygrothermal performance of retrofitted timber frame assemblies in six us climates.” In *ASCE International Conference on Computing in Civil Engineering*. Corvallis, Oregon. June 25 - 28.

Recart, C., **Méndez Echenagucia, T.**, and Sturts Dossick, C. (2024). “Hygrothermal performance of energy retrofits in accordance with the current energy code across three climate zones in the u.s.” In *uSIM2024 -Shaping Net Zero Policies with Building Simulation - The 4th IBPSA-Scotland Conference*. Edinburgh.

Wiggins, M., Woo Lee, H., and **Méndez Echenagucia, T.** (2024). “Concurrent modeling of embodied carbon and construction costs for mass timber construction.” In *Construction Research Congress 2024 - Constructing Connections across Research and Industry*. Des Moines - Iowa.

Salesin, J., Lokki, T., and **Méndez Echenagucia, T.** (2025). “Seat sublime: Focusing on the totality of the concertgoer experience – acoustical, visual, and physical.” *Proceedings of Meetings on Acoustics*, 58(1):015001. ISSN 1939-800X.

Book chapters

Filomeno Coelho, R., **Méndez Echenagucia, T.**, Pugnale, A., and Richardson, J.N. (2014). “Genetic algorithms for structural design.” In S. Adriaenssens, P. Block, D. Veenendaal, and C. Williams, editors, *Shell Structures for Architecture - Form finding and optimization*, chapter Apendix C, pages 290–294. Routledge, London.

Pugnale, A., **Méndez Echenagucia, T.**, and Sassone, M. (2014). “Computational Morphogenesis - Design of a free-form concrete shell.” In S. Adrianssens, P. Block, D. Veenendaal, and C. Williams, editors, *Shell Structures for Architecture - Form finding and optimization*, chapter 18, pages 224–236. Routledge, London.

Hebel, D., Block, P., Heisel, F., and **Méndez Echenagucia, T.** (2017). “Beyond mining - urban growth: The architectural innovation of cultivated resources through appropriate engineering.” In A.Z.P. Hyungmin Pai, editor, *Imminent Commons: The Expanded City*. Actar Publishers. Seoul Biennale of Architecture and Urbanism 2017.

Van Mele, T., **Méndez Echenagucia, T.**, Pigram, D., Liew, A., and Block, P. (2021). "Ultralight formwork system for thin, textile-reinforced concrete shells." In J. Schoof, editor, *Robust Resilient Resistant: Reinforced Concrete Structures*, pages 20–28. Edition DETAIL, Munich.

Anik, M.S. and **Méndez Echenagucia, T.** (2023). "Efficient parametric design-space exploration with reinforcement learning recommenders." In N. Abbasabadi, editor, *Artificial Intelligence in Performance-Driven Design: Theories, Methods, and Tools*. Wiley.

Peer-reviewed conferences without proceedings

Méndez Echenagucia, T. (2010a). "Herramientas informáticas para la investigación y proyectos en el campo de la arquitectura y construcción: Caso de estudio, sistema de parasoles en madera de teca para fachadas de edificios de mediana altura." In *XXVII Jornadas de Investigación del IDEC*. July 6 - 8.

Méndez Echenagucia, T. (2010b). "Sistema de parasoles en madera de teca para fachadas de edificios de mediana altura: Caso de estudio, edificios de oficina en caracas." In *XXVII Jornadas de Investigación del IDEC*. July 6 - 8.

Shtrepi, L., Menichelli, J., Astolfi, A., **Mendez Echenagucia, T.**, and Masoero, M.C. (2017). "Improving scattering surface design with rapid feedback by integrating parametric models and acoustic simulation." In *Acoustic Society of America 2017 Annual Meeting*. New Orleans - Louisiana. December 4 - 8.

Rajendran, V., **Méndez Echenagucia, T.**, and Piacsek, A. (2020). "Design of efficient low-frequency sound absorbers using an array of helmholtz resonators." In *Acoustic Society of America 2020 Annual Meeting*. Virtual conference. December 7 - 11.

Brown, N. and **Méndez Echenagucia, T.** (2021). "Minimizing radiated sound power through geometric stiffness in mass timber floor systems." In *Acoustic Society of America 2021 Annual Meeting*. Seattle - Washington. November 29 December 3.

Rajendran, V., Piacsek, A., and **Méndez Echenagucia, T.** (2021). "Design of broadband helmholtz resonator arrays using the radiation impedance method." In *Acoustic Society of America 2021 Annual Meeting*. Seattle - Washington. November 29 December 3.

Méndez Echenagucia, T., Piacsek, A., and Roozen, N. (2024). "Cross laminated timber geometry and joinery implications on low-frequency sound transmission: Fea and laser doppler vibrometry studies." In *Acoustic Society of America 2024 Annual Meeting*. Ottawa - Canada.

Iyer, H., Tabatabaei Manesh, M., Kim, R., Dillon, M., Kim, D., **Méndez Echenagucia, T.**, and Roumeli, E. (2025). “Algal biomatter foams for acoustic insulation applications.” In *Materials Research Society Spring Meeting*. Seattle - Washington.

Tabatabaei Manesh, M., Iyer, H., Kim, R., Dillon, M., Kim, D., **Méndez Echenagucia, T.**, and Roumeli, E. (2025). “Broadband sound absorption via algal biofoams and helmholtz resonator arrays.” In *Acoustic Society of America 2025 Annual Meeting joint with the 25th International Congress on Acoustics*. New Orleans - Louisiana.

Tabatabaei Manesh, M. and **Méndez Echenagucia, T.** (2025a). “Design and optimization of meta-material ventilated sound barriers for building facades: An experimental and numerical investigation.” In *Acoustic Society of America 2025 Annual Meeting joint with the 25th International Congress on Acoustics*. New Orleans - Louisiana.

Tabatabaei Manesh, M. and **Méndez Echenagucia, T.** (2025b). “A data-driven framework for predicting the acoustic performance of metamaterial arrays.” In *Joint Meeting Acoustical Society of America and Acoustical Society of Japan*. Honolulu - Hawaii.

Méndez Echenagucia, T. and Tabatabaei Manesh, M. (2025). “Surrogate modelling of the vibro-acoustic behavior of cross-laminated timber floor slabs.” In *Joint Meeting Acoustical Society of America and Acoustical Society of Japan*. Honolulu - Hawaii.

Conference posters

Méndez Echenagucia, T., Shtrepi, L., Badino, E., and Astolfi, A. (2019). “Investigating the importance of geometrical accuracy in acoustic simulations: A comparison of NURBS and mesh-based approaches.”

Bjarvin, C., Pierobon, F., **Méndez Echenagucia, T.**, and Ganguly, I. (2023). “Reuse, recycle, incinerate or landfill? lca-based environmental implications of end-of-life scenarios for mass timber buildings.” Conference Poster - Mass Timber conference, Portland Oregon 2023.

Brown, N., **Méndez Echenagucia, T.**, Sprague, T., Lohr, J., Di Furia, R., and Hunter, J. (2023). “Mass timber joinery design for digital fabrication and de-constructability.” Conference Poster - Mass Timber conference, Portland Oregon 2023.

Droog, L., Pierobon, F., Huang, M., Carlisle, S., **Méndez Echenagucia, T.**, Simonen, K., and Ganguly, I. (2023). “Parametric open data for life cycle assessment.” Conference Poster - Mass Timber conference, Portland Oregon 2023.

Other publications

Van Mele, T., **Méndez Echenagucia, T.**, Pigram, D., Liew, A., and Block, P. (2018). "A prototype of a thin, textile-reinforced concrete shell built using a novel, ultra-lightweight, flexible formwork system." DETAIL.

Moroseos, T., Meek, C., and **Méndez Echenagucia, T.** (2024). "The carbon sweet spot: Design tradeoffs for embodied and operational carbon in new buildings." Urban Land Institute - Report. ULI.

Méndez Echenagucia, T. and Cheng, R. (2024). "A call for organized open-source software infrastructure for the aec industry." Practice Management Digest - American Institute of Architects. AIA.

II.B Research projects and prototypes

This section contains research projects, prototypes and lab experiments I have been involved in as an Assistant Professor at the University of Washington, Postdoctoral Researcher at the Block Research Group in ETH Zürich, and a Doctoral student at the Politecnico di Torino:

1- Assistant Professor - University of Washington

2022 - 2023 **Mass timber floor system prototypes and LDV measurements**
with Nathan Brown (UW), Bert Roozen (KU Leuven) and Andy Piacsek (CWU)
Role: PI
Design, fabrication and assembly of two CLT floor prototypes on a Glulam frame
Laser Doppler Vibrometry measurements for acoustic vibro-acoustic performance
Seattle - WA, USA
Sample publication: **Méndez Echenagucia, T.** et al. (2024)
UW CBE Spotlight: Plywood on steroids

2020 **Helmholtz resonator array prototypes**
with Vidhya Rajendran (UW), NBBJ and Arup acoustics
Role: co-PI
Design, fabrication and on-site testing of resonant acoustic absorption panels in Cross Laminated Timber
Seattle - WA, USA
Sample publication: Rajendran et al. (2022)
Popular Media: Fast Company, Input

2- Postdoctoral researcher - Block Research Group (BRG), ETH Zürich

2015 - 2019	**NEST HiLo** with BRG, Architecture and Building Systems Group, ROK Architects, Buergin Creations, Dr. Schwartz Consulting AG and Marti AG Role: BRG co-lead on the concrete shell roof and fabric formwork system Design and construction of a flexible workspace unit for the NEST building Structural design, system design and fabrication for a concrete sandwich shell roof built with a cable-net and formwork system Dübendorf, Switzerland Sample publication: Block et al. (2017) Popular Media: Designboom
2017	**Acoustic testing of funicular floor system** with Bert Roozen (KU Leuven) Role: PI Large scale prototype of funicular floor system for vibroacoustic lab testing Structural design and digital fabrication of floor sample Sample publication: **Méndez Echenagucia, T.** and Roozen and Block (2016)
2017	**Cable-net and fabric formwork prototype** with BRG, supermanouvre, Marti AG and Bollinger Grohman Role: BRG co-lead Large scale prototype for the Cable-net and fabric formwork for the HiLo roof Structural design and digital fabrication of 1:1 scale thin shell concrete prototype On-site supervision of fabricators and general contractor Sample publication: **Méndez Echenagucia, T.** et al. (2019) Popular Media (selected): Archdaily, BBC News, Architect Magazine
2017	**Mycotree** with BRG, chair of Sustainable Construction (KIT) Role: Contributor on structural design and fabrication Design of a compression only branching structure in mycelium for the Seoul Biennale Seoul, South Korea Sample publication: Hebel et al. (2017) Popular Media (selected): Dezeen, Detail, Word-Architects
2015 - 2016	**Beyond Bending** with BRG, John Ochsendorf, ODB Engineering, and Escobedo construction Role: Contributor on structural design and fabrication Exhibition at the Venice Biennale 2016 Structural design and construction of a 15m span compression only vault in sandstone Structural design and fabrication of a concrete funicular floor system prototype Venice, Italy Sample publication: Rippmann et al. (2016) Popular Media (selected): Wired, Forbes, Domus

2015 - 2016 **Droneport Prototype**
with BRG, Norman Foster Foundation, EPFL, ODB, Universidad Politécnica de Madrid
Role: Contributor on structural design and construction
Exhibition at the Venice Biennale 2016, Italy
Structural Design and construction of a thin tile vault pavilion
Popular Media (selected): L'Architecture d'Aujourd'hui , Dezeen, Architects Journal

2015 **ETH Pavillion**
with BRG, chair of Sustainable Construction (KIT)
Role: BRG Lead
Ideas City Festival New York City, USA
Design and construction of a pavilion in recycled materials for the "Ideas City Festival"
New York City, USA
Popular Media (selected): Dezeen, Azure

2014 - 2015 **Fábrica de Cultura**
with BRG, Urban Think Tank (ETH), Universidad de Norte Barranquilla
Role: BRG co-lead
Structural design and acoustic analysis of a vaulted auditorium for a public arts school
Sample publication: **Méndez Echenagucia, T.** and Block (2015)
Barranquilla, Colombia

3- Ph.D. Student - Politecnico di Torino

2012 **Summer studio pavilions**
Role: Studio instructor
Design and construction of three pavilions built with students during a summer studio
Pavilions made in on-site concrete, tile vaulting and timber frames
Turin, Italy

2012 **Brickshell**
Role: Studio instructor
Design and construction of tile vaulted shell built with students during a summer studio
Sample publication: Rajabzadeh et al. (2014)
Turin, Italy

2011 **Monalisa pavillion**
Role: Studio instructor
Design and construction of a timber pavilion for the MADE exhibition
Milan, Italy

II.C Open-source software development

2024 - present **POD / LCA**
University of Washington
Python library and tools ecosystem for Dynamic Lifecycle Analysis
Role: Lead developer
GitHub repository

2022 - 2024 **COMPAS Energy+**
University of Washington
Python library for room building energy performance simulation based using Energy+
Role: Lead developer

2021 - 2022 **UpStream**
University of Washington
with Chuou Zhang, Indroneil Ganguly (UW), Marty Brennan, Jacob Dunn (ZGF)
Role: Faculty supervisor with Prof. Indroneil Ganguly (UW)
UpStream Forestry Carbon and LCA Tool
Published in: Metropolis
GitHub repository

2020 - 2024 **COMPAS Room Acoustics**
University of Washington
Role: Lead developer
Python library for room acoustics simulation based on the COMPAS framework

2019 - 2024 **COMPAS Helmholtz**
University of Washington
Role: Lead developer
Python library for the design of acoustic resonant absorption panels

2018 - present **COMPAS Vibro**
University of Washington / ETH Zürich
Role: Lead developer
Python library for vibro-acoustic simulation based on the Rayleigh integral
GitHub repository

2016 - 2020 **COMPAS FEA**
University of Washington / ETH Zürich / University of Sheffield
Role: Co-lead developer
Python library for finite element analysis of building structures
GitHub repository

2015 - 2019 **COMPAS**
ETH Zürich
Computational framework for Architecture and Structures
Role: Co-developer
GitHub repository

II.D Grants, fellowships and third-party funding

2024	**NSF - University of Washington** Can Irregular Structural Patterns Beat Perfect Lattices? Biomimicry for Optimal Acoustic Absorption Role: Co-Principal Investigator with: Prof. Ed Habtour (UW AA), Mostafa Nouh (University at Buffalo) Awarded funds: $ 293,331.00
2023	**UW Population Health initiative Climate Change Pilot Grant - University of Washington** Sustainable metamaterials for insulation applications Role: Co-Principal Investigator with: Prof. Eleftheria Roumeli (UW MSE) Awarded funds: $ 49,356
2023	**UW Population Health initiative planning grant - University of Washington** Sustainable materials for structural applications Role: Co-Principal Investigator with: Prof. Eleftheria Roumeli (UW MSE), Kate Simonen (UW Arch.) Awarded funds: $ 10,000
2022	**ARPA-e - University of Washington** Parametric Open Data for Life Cycle Assessment (POD — LCA) Role: Co-Investigator with: Kate Simonen, Chris Meek (UW Arch.), Indroneil Ganguli (UW CE) Awarded funds: $ 3,744,303
2020	**Royalty Research Fund - University of Washington** UW Office of Research internal grant to faculty seeking to establish new research Role: Principal Investigator Awarded funds: $ 39.500
2020	**Arup Acoustics in-kind contribution** Expert advice and on-site measurements of the acoustic performance of a Helmholtz resonator array Role: Principal Investigator In kind contribution: $ 16.500
2020	**NBBJ Design** Funding of a research assistant for summer position Role: Principal Investigator Funded time: 2 months, 20 hours per week

2015 - 2018	**HiLo Industry funding** BRG - ETH Zürich Role: BRG Lead on roof Secured funding of the HiLo project from Holcim Switzerland, BASF, Autodesk and Doka
2017	**European Research Council Synergy grant** (Unsuccessful) ETH Zürich with Vrije Universiteit Brussel and University of Patras Role: Contributor Application for research on reducing the carbon emissions of floor slabs Requested funds: € 9.664.773
2016	**Beyond Bending Industry funding** ETH Zürich Role: Contributor Securing project sponsoring from the Escobedo Group, Artemide and Pro Helvetia
2015 - 2016	**ETH Foundation** Post-doctoral research grant. Research on sound insulation in the low frequencies by means of structural stiffness Role: Principal Investigator Awarded funds: CHF 100.000
2011 - 2013	**Italian Ministry of Higher Education and Research (MIUR)** PhD scholarship at the Politecnico di Torino Role: Recipient Awarded stipend and tuition fees for 3 years

II.E Invited lectures and Panels

2026	**European Acoustics Association - Summer School** TU Graz - Austria Lecture and dicussion "Design and analysis of porous and resonant absorbers"
2026	**Confluence Seattle** Seattle - WA Panel discussion "Beyond Generative Design: AI and the Next Phase of Intelligence in the Built Environment"
2026	**Westside Acoustics Research Week** (remote) Los Angeles - CA Research presentation "Form Over Mass: Using Geometry to Control Low-Frequency Noise in Low-Carbon Floor Systems"
2026	**Architecture in the Age of AI** Seattle - WA Panel discussion, Organizer and Moderator University of Washington, Department of Architecture

2025	**Vancity Computational Design** (remote) Vancouver - Canada Fast and Epp's Concept Lab "Digital fabrication for Multi-Objective Building Components"
2025	**Making Through Technology** Seattle International Architecture Forum AIA Seattle - LMN Architects "Digital fabrication for Multi-Objective Building Components"
2024	**Computer Science for the Environment** Lunch talk - Seattle University of Washington - eScience Institute "Computational design for sustainable buildings"
2023	**Council on Tall Buildings and Urban Habitat Americas** Conference - Seattle Panel discussion "Decarbonizing Design – Material and Technical Advancements"
2023	**Passive House Northwest** Conference - Portland, Oregon Keynote Speech "Computational geometry for sustainable construction"
2023	**University of Nebraska** - Lincoln, Nebraska Guest lecture for large undergrad course on computational methods "Computational geometry for sustainable construction"
2023	**Carbon Leadership Forum Portland** (remote) February meeting Presentation on research studio and Operational + Embodied carbon research "Designing for the triple bottom line: Experience, Energy and Embodied Carbon Emissions"
2022	**Society of Building Science Educators** (SBSE) retreat Presentation on research studio teaching "Carbon Research Studio 2021: Designing for low emissions"
2022	**University of Melbourne** (remote) Panel discussion after a presentation from Prof. Trevor Cox ABP Symposium 2022 - Melbourne Australia
2022	**Kent State University** (remote) As part of a workshop on Pachyderm acoustics taught by Arthur Van der Harten "Computational methods for room acoustic design"
2021	**Committee of 33** (remote) Non-profit organization in Seattle - WA "Designing for low emisions"

2021	**Associazione Italiana di Acustica** (remote) "Aspettando Matera" live streamed presentation series "Progettare con/le onde: Performance based design per la progettazione acustica in architettura"
2021	**University College London** (remote) Bartlett Faculty of the Built Environment "Computational geometry for sustainable construction"
2019	**Politecnico di Torino** - Turin, Italy Dipartimento di Architettura e Design "Computational geometry for sustainable construction"
2019	**University of Toronto** - Toronto, Canada Daniels Faculty of Architecture, Landscape, and Design "Computational geometry for sustainable construction"
2019	**Université de Montréal** - Montreal, Canada Faculté de l'Aménagement "Computational geometry for sustainable construction"
2019	**Université Catholique de Louvain** - Louvain-la-Neuve, Belgium Faculté d'ingénierie architecturale "Computational geometry for sustainable construction"
2019	**University of Washington** - Seattle, United States of America Department of Architecture "Computational geometry for sustainable construction"
2018	**Universidad de Granada** - Granada, Spain E.T.S. de Arquitectura "Geometría de estructuras sostenibles"
2018	**Architekturzentrum Wien** - Vienna, Austria PechaKucha for Concrete manufacturing consortium "Funicular floor systems"
2017	**EMPA Nest** - Zurich, Switzerland Lafarge Holcim Sustainable construction "The HiLo research unit"
2017	**KU Leuven** - Leuven, Belgium Division Acoustics and Thermal Physics "Acoustic insulation through structural stiffness"

2017	**Hochschule Ostwestfalen-Lippe** - Detmold, Germany Master of Engineering in Integrated Design "Computational search in architectural design"
2016	**Università IUAV di Venezia** - Venice, Italy with Prof. Philippe Block "Strength through geometry"
2016	**Heatherwick Studio** - London, England Architecture firm with Prof. Philippe Block "Strength through geometry"
2015	**Massachusetts Institute of Technology** - Boston, United States of America Building Technology Group "Computational search in architectural design"
2015	**University College London** - London, England The Bartlett School of Architecture "Search algorithms for acoustic design"
2009	**Universidad Simón Bolívar** - Caracas, Venezuela Departamento de Diseño, Arquitectura y Artes Plásticas "Forma acústica arquitectónica y esctructural"

II.F Peer reviewing

- Natural Sciences and Engineering Research Council of Canada (NSERC)
- International Association for Shells and Spatial Structures (IASS)
- Symposium on Simulation for Architecture and Urban Design (SimAUD)
- University of Washington Royalty Research Fund (RRF)
- Applied Acoustics
- Symposium on Computational Fabrication 2022
- Forum Acusticum 2023
- Energy Science & Engineering
- Computers & Structures

II.G Conference session chairing

2022 **Symposium on Computational Fabrication**
Architecture section
Chair

2018 **IASS Boston**
Shell structures
Co-chair with Tim Michiels

2015 **IASS Amsterdam**
Performance Aided Design
Co-chair with Prof. Dario Parigi

III Architectural Practice

III.A Projects

2023 **St.Joseph children's home** (Concept design)
Jinja, Uganda
Design for a temporary children's home
with Tomás Mena

2020 **Arena design competition** (Unbuilt)
Vienna, Austria
Competition consultant in structural design and acoustics for Mass Timber Arena
with Zaha Hadid Architects, Blumer Lehmann AG, Arup

2014 **Piazza Bernini** (Unbuilt)
Turin, Italy
Residential development in the city center
with Frlan + Jansen Architetti
Architectural Design

2013 **Reale Mutua** (Unbuilt)
Turin, Italy
Invited competition for an Office building in the city center
with Frlan + Jansen Architetti
Architectural Design

2013 **Teatro Colón** (Unbuilt)
Bogotá, Colombia
Competition for an addition to a historical Theater
with ADJKM Arquitectos and Marco Palma
Concert Hall Design, Parametric Modeling and Geometry consultancy

2013	**Lacacao** (Unbuilt) Boa Vista island, Capo Verde Preliminary design for a beach resort and residence with Frlan + Jansen Architetti Architectural Design
2012 - 2013	**Via Nizza** Turin, Italy Refurbishment an historical building for residential development with Frlan + Jansen Architetti Architectural Design
2011 - 2013	**CASMSB** (Unbuilt) Caracas, Venezuela Music conservatory and concert hall, Venezuela Concert Hall Design Parametric Modelling / computational geometry consultancy
2012	**Holcom Headquarters** Beirut - Lebanon Corporate office Building envelope & shading consultancy Parametric Modelling
2011	**Lavazza Headquarters** (Unbuilt) Turin - Italy Invited competition for Corporate offices Architectural Design Parametric Modelling
2011	**Strada Andezeno** Chieri, Italy Mixed housing/commercial building with Frlan + Jansen Architetti Architectural Design
2011	**Kleinste Soep Fabriek** (Unbuilt) Groningen, the Neatherlands Entrance Pavilion and Foyer for a soup Factory with Frlan + Jansen Architetti Architectural Design
2011	**Sestriere** Sestriere, Italy Ski residence with Frlan + Jansen Architetti Geometry and Material Optimization consultancy

2011	**Santa Clara** Chieri, Italy Residential development in an ex-consecrated church with Frlan + Jansen Architetti Architectural Design
2011	**Venice city Visions** Venice, Italy Ideas competition on the future of Venice with Claudio Sframelli, Jacopo Bracco and Alberto Lessan Architectural Design
2011	**Venice city Visions** Turin, Italy Private Loft Photography studio and exhibition space with Frlan + Jansen Architetti Architectural Design
2010	**CASMSB competition** Caracas, Venezuela Music Education complex competition TMS Arquitectura Architectural Design
2010	**EIM** (Unbuilt) La Guaira, Venezuela Shopping center TMS Arquitectura Solar shading and design consultancy
2010	**Stonehill Masterplan** Bangalore, India Mixed urban development, housing, office space, and cultural center with Frlan + Jansen Architetti Architectural Design
2010	**Residency Road** Bangalore, India Private Office Building with Frlan + Jansen Architetti Architectural Design
2010	**Oficina La Lagunita** Caracas, Venezuela Private offices interiors for an investment group TMS Arquitectura Architectural Design

2010 **Centro Simon Díaz competition** (Unbuilt)
Caracas, Venezuela
Cultural Center in an informal settlement in Caracas
TMS Arquitectura
Architectural Design

2010 **Hangar** (Unbuilt)
Ocumare, Venezuela
Private hangar in a Civil Airport
TMS Arquitectura
Architectural Design

2009 **Bima** (Unbuilt)
Caracas, Venezuela
Invited competition for a retail Store for a Furniture and design brand
TMS Arquitectura
Architectural Design

2009 **Campo Plaza**
Caracas, Venezuela
Private Apartment Refurbishment
TMS Arquitectura
Architectural Design

2009 **Campo Norte**
Caracas, Venezuela
4 Private Apartment Refurbishments in Caracas
TMS Arquitectura
Architectural Design

2009 **Centro Empresarial Boleita** (Unbuilt)
Caracas, Venezuela
Mixed Office/Commercial building
TMS Arquitectura
Architectural Design

2008 **Chieri Services Center** (Unbuilt)
Chieri, Italy
Mixed use commercial/Hotel and public building
with Frlan + Jansen Architetti
Architectural Design

2008 **Zona Rental** (Unbuilt)
Caracas, Venezuela
Mixed Use commercial / cultural complex competition
TMS Arquitectura + Miguel Acosta and Alfonso Paolini
Architectural Design

2008	**Las Perezas** (Unbuilt) Caracas, Venezuela Private Residence TMS Arquitectura Architectural Design
2008	**Superagencia** Caracas, Venezuela Refurbishment of private offices for a Design firm TMS Arquitectura Architectural Design
2008	**Polar La Yaguara** Caracas, Venezuela Medical station for a Manufacturing Plant TMS Arquitectura Architectural Design
2008	**Polar La Guaira** La Guaira, Venezuela Office expansion and Medical station for a Distribution center TMS Arquitectura Architectural Design
2008	**Polar Ocumare** Ocumare, Venezuela Storage Building for a distribution center TMS Arquitectura Architectural Design
2007	**Clínica Plaza** (Unbuilt) Caracas, Venezuela Refurbishment and extension of a private Health Care facility TMS Arquitectura Architectural Design
2007	**Saltarrana** Caracas, Venezuela Private Residence with AM: Arquitectura Multimedia Architectural Design
2007	**Perugia** (Unbuilt) Perugia, Italy Student Housing competition with Frlan + Jansen Architetti Architectural Design

2007	**Casa Facta** Perinera, Italy Private sky residence in the Italian Alps with Frlan + Jansen Architetti Architectural Design
2006	**Falconara Market** (Unbuilt) Falconara Maritima,Italy Open air market Competition (1st Prize) with Frlan + Jansen Architetti Architectural Design
2004	**Casa Wenzelmann** (Unbuilt) Caracas, Venezuela Private residence Architectural Design

IV Teaching

IV.A University of Washington courses

Spring 2026	498 B /598 B	**AI of Architectural Design** Introduction to Artificial intelligence methods and libraries for early stage architectural design	3 credits
Spring 2026	Arch 436A/436B	**Building Acoustics** Applied acoustics course on room acoustics and sound insulation in buildings	3 credits
Winter 2026	498 C /598 D	**Computational design** Applied computational design course based on the Python programming language	3 credits
Autumn 2025	Arch 586	**Computation and design technology seminar** Seminar on the use of computational methods for architectural design technology	3 credits
Spring 2025	Arch 436A/598C	**Building Acoustics** Applied acoustics course on room acoustics and sound insulation in buildings	3 credits
Spring 2025	Arch 592 A	**Research methods** an overview of the role and practice of research methods in architecture Co-taught with Prof. Brian McLaren	3 credits

Spring 2024	Arch 600	**Independent study - Mohammad Tabatabai** Ventilated sound barriers - Numerical models and parametric studies	5 credits
Spring 2024	Arch 592 A	**Research methods** an overview of the role and practice of research methods in architecture Co-taught with Prof. Brian McLaren	3 credits
Winter 2024	Arch 600	**Independent study - Mohammad Tabatabai** Ventilated sound barriers - Numerical models and parametric studies	5 credits
Winter 2024	498 B /598 C	**Computational design** Applied computational design course based on the Python programming language	3 credits
Autumn 2023	Arch 600	**Independent study - Mohammad Tabatabai** Ventilated sound barriers - Numerical models and parametric studies	5 credits
Autumn 2023	Arch 523	**Design Technology IV** Integration of structure, environmental systems, spatial organization and architectural form Co-taught with Prof. Rob Peña	3 credits
Autumn 2023	Arch 586	**Computation and design technology seminar** Seminar on the use of computational methods for architectural design technology	3 credits
Autumn 2023	Arch 600	**Independent study - Mohammad Tabatabai** Ventilated sound barriers - Literature review and analytical modeling	5 credits
Spring 2023	Arch 508 B	**Research Studio II** Studio on the trade-offs between operational and embodied emissions of mass-timber housing buildings Co-taught with Prof. Chris Meek	6 credits
Spring 2023	Arch 594 A	**Research Studio II Seminar** Seminar on computational methods for the estimation carbon emissions Co-taught with Prof. Chris Meek	3 credits
Winter 2023	498 B /598 C	**Computational design** Applied computational design course based on the Python programming language	3 credits

Autumn 2022	Arch 598 A	**Computation and design technology seminar** Seminar on the use of computational methods for architectural design technology	3 credits
Autumn 2022	Arch 600	**Independent study - Nathan Brown** Finite Element Analysis of Timber joinery	3 credits
Spring 2022	Arch 592 A	**Research methods** an overview of the role and practice of research methods in architecture Co-taught with Prof. Brian McLaren	3 credits
Spring 2022	598 C	**Computational design** Applied computational design course based on the Python programming language	3 credits
Winter 2022	Arch 598C	**Computation and design technology seminar** Seminar on the use of computational methods for architectural design technology Newly developed	3 credits
Autumn 2021	Arch 600	**Independent study - Nathan Brown** Documentation and presentation of vibro-acoustic studies for mass timer floors	3 credits
Autumn 2021	Arch 523	**Design Technology IV** Integration of structure, environmental systems, spatial organization and architectural form Co-taught with Prof. Rob Peña	3 credits
Spring 2021	Arch 508 B	**Research Studio II** Studio on the trade-offs between operational and embodied emissions of mass-timber buildings Co-taught with Prof. Chris Meek	6 credits
Spring 2021	Arch 594 B	**Research Studio II Seminar** Seminar on computational methods for the estimation carbon emissions Co-taught with Prof. Chris Meek	3 credits
Winter 2021	Arch 587 A	**Theory of design computing** Seminar on the theory and practice of contemporary computational methods	3 credits

Autumn 2020	Arch 523	**Design Technology IV** Integration of structure, environmental systems, spatial organization and architectural form Co-taught with Prof. Rob Peña	3 credits
Spring 2020	Arch 592 A	**Research methods** an overview of the role and practice of research methods in architecture Co-taught with Prof. Brian McLaren	3 credits
Spring 2020	Arch 498 / 598 C	**Computational design** Applied computational design course based on the Python programming language Newly developed	3 credits
Winter 2020	Arch 587 A	**Theory of design computing** Seminar on the theory and practice of contemporary computational methods	3 credits
Autumn 2019	Arch 523	**Design Technology IV** Integration of structure, environmental systems, spatial organization and architectural form Re-developed Co-taught with Prof. Rob Peña	3 credits

IV.B ETH Zürich courses

Courses taught at ETH Zürich while working as a post-doctoral researcher. The courses were officially offered by Professors Philippe Block and Joseph Schwartz.

Spring 2019		**Structural Design VI** State-of-the-art computational structural design Taught with Dr. Matthias Rippmann	3 credits
Spring 2018		**Structural Design VI** State-of-the-art computational structural design Taught with Dr. Matthias Rippmann	3 credits
Spring 2016		**Structural Design (MAS)** Introduction to computational structural design for Master of Applied Science in Digital Fabrication Taught with Dr. Matthias Rippmann	3 credits

IV.C Politecnico di Torino courses

Summer courses taught as a Ph.D. student with Prof. Mario Sassone.

Summer 2014	**Computational Design** Computational design summer workshop Taught with Prof. Mario Sassone
Summer 2013	**Computational Design** Computational design summer workshop Taught with Prof. Mario Sassone
Summer 2012	**Computational Design** Computational design summer workshop Taught with Prof. Mario Sassone

IV.D Student teaching evaluation scores

Quarter	Course	Respond / Enrolled	Median	Adj. Median	COVID
Spring 2026	Arch 436A/436B	14 / 33	4.7	4.7	In person
Spring 2026	Arch 498B/598B	6 / 115	4.6	4.2	In person
Winter 2026	Arch 498C/598D	15 / 27	4.6	4.6	In person
Autumn 2025	Arch 586	8 / 17	4.8	4.4	In person
Spring 2025	Arch 436/598C	8 / 27	4.6	4.4	In person
Spring 2024	Arch 592A	16 / 34	4.8	4.7	In person
Winter 2024	Arch 498B / 598C	20 / 33	4.6	4.7	In person
Autumn 2023	Arch 523	20 / 47	4.8	4.7	In person
Autumn 2023	Arch 586	6 / 8	4.7	4.3	In person
Spring 2023	Arch 508 B	5 / 8	5.0	4.4	In person
Spring 2023	Arch 594 A	6 / 8	4.9	4.3	In person
Winter 2023	Arch 498B/598 C	15 / 27	4.7	4.7	In person
Autumn 2022	Arch 598 A	7 / 10	4.6	4.2	In person
Spring 2022	Arch 598 C	11 / 16	4.7	4.7	Hybrid

Spring 2022	Arch 592 A	19 / 43	4.2	4.2	Hybrid
Winter 2022	Arch 598 C	10 / 14	4.9	4.6	Hybrid
Autumn 2021	Arch 523	27 / 53	4.4	4.7	Hybrid
Spring 2021	Arch 594 B	7 / 16	4.6	4.3	Remote
Spring 2021	Arch 508 B	6 / 16	4.9	4.5	Remote
Winter 2021	Arch 587	10 / 14	4.8	4.7	Remote
Autumn 2020	Arch 523	24 / 44	4.6	4.7	Remote
Spring 2020	Arch 498 - 598 C	13 / 23	4.6	4.7	Remote
Spring 2020	Arch 592	23 / 32	4.2	4.2	Partially remote
Winter 2020	Arch 587	9 / 13	4.8	4.7	No
Autumn 2019	Arch 523	33 / 46	4.2	4.3	No

IV.E Post-doctoral Supervision

2024 - ongoing **Isuru Nanayakkara**
University of Washington - Department of Architecture
Project: Python programming and tools development for the POD-LCA project
Stress field analysis and optimization for timber joinery

IV.F Ph.D. Supervision

2023 - ongoing **Mohammad Tabatabai** - Chair
University of Washington - College of Built Environments
Title: TBD

2025 - 2026 **Hareesh Iyer**- Committee member
University of Washington - Materials Science and Engineering
Title: TBD

2023 - 2025 **Benjamin Tod Jones** - Graduate School Representative
University of Washington - Computer Science and Engineering
Title: "Neural Duals for Symbolic CAD"

2021 - 2023 **Carolina Recart** - Committee member
University of Washington - Civil And Environmental Engineering

Title: Hygrothermal performance of energy retrofits in buildings. An assessment of the residential building stock in the US

2021 - ongoing **Hannah Twigg-Smith** - Graduate School Representative
University of Washington - Human Centered Design and Engineering
Title: TBD

2021 - 2024 **Sarah Wichman** - Graduate School Representative
University of Washington - Civil And Environmental Engineering
Title: Seismic Behavior of Tall Rocking Mass Timber Walls

2020 **Blanca Perez** - Host
Visiting PhD student - Universidad Politécnica de Valencia, Spain
Topic: Parametric modeling of concert hall acoustics and materials

IV.G Ph.D. Supervision as Post-doctoral researcher at ETH Zürich

While holding the position of Post-doctoral researcher, contributed to the supervision of the PhD candidates at the Block Research Group, Institute for Technology in Architecture ETH Zürich. Providing constant feedback on their research topics, imparting instruction in Python programming, computational design, computational geometry and structural design.

2018 — 2019 **Alessandro Dell'Endice**
Assessment methods for historical masonry structures

2017 — 2019 **Crstián Calvo Barentin**
Design and fabrication of funicular floor systems for low environmental impact

2015 — 2018 **Juney Lee**
"Computational Design Framework for 3D Graphic Statics"

2014 — 2019 **David López López**
"Tile vaults as integrated formwork for concrete shells"

IV.H Thesis Supervision at the University of Washington

2025 - 2026 **Shafiul Islam**
Role: Chair
Master of Science of Architecture - Design Computing
Graph based autoregressive machine learning for high-performance residential layout generation
Committee: Mehlika Inanici, Mohammad Tabatabai

2025 - 2026 **Hasibul Hossain**
Role: Chair
Master of Science of Architecture - Design Computing
Fluid Intelligence: Physics-informed neural networks as CFD surrogates for indoor airflow and thermal confort
Committee: Narjes Abbasabadi, Mohammad Tabatabai

2024 - 2025 **Eva Zaphie Zhang**
Role: Committee member
Master of Science of Architecture - Design Technology
Leveraging Machine Learning to Uncover Integrated Impacts of Urban Environment Features on Human Health
Committee: Narjes Abbasabadi, Edmund Seto

2024 - 2025 **Ethan Matthew Cano**
Role: Committee member
Master of Architecture
TBD
Committee: Narjes Abbasabadi, Vince Wang

2023 - 2024 **Jeremy Salesin**
Role: Chair
Master of Architecture
Seat Sublime: A Recommendation System to Accommodate User Preference in Concert Hall Seating Selection
Committee: Rob Peña

2022 - 2023 **Md. Shariful Anik**
Role: Chair
Master of Science of Architecture - Design Technology
AI-Driven design exploration: Use of recommender system in parametric design-space exploration
Committee: Narjes Abbasadabi, Karthik Mohan

2022 - 2023 **Nathan Brown**
Role: Chair
Master of Science of Architecture - Design Technology
Mass Timber Joinery Design for Digital Fabrication and De-constructability
Committee: Tyler Sprague

2022 - 2023 **Steven Youn**
Role: Committee member
Master of Urban Planning
Improving Pedestrian-Friendly Building-Street Relationships through Computational Design towards Seattle
Chair: Daniel Abramson

2021-2022	**Preston Pape** Role: Committee member Master of Science of Architecture - Design Technology Title: Metamodeling of Energy and Operational Carbon in Detached Accessory Dwelling Units Chair: Rick Mohler
2021-2022	**Christina Bjarvin** Role: Committee member Master of Science - School of Environmental & Forest Sciences Title: Assessing the Carbon Balance for Mass Timbers Beyond the First Life Chair: Indroneil Ganguly
2021	**Matt Wiggins** Role: Committee member Master of Science in Construction Management Cost model for embodied carbon reduction: comparative study between mass timber and structural steel Chair: Hyun Woo Lee
2020 - 2021	**James Blanchard** Role: Chair Master of Architecture Acoustic spatial design methodology and its application in health environments Committee: Heather Burpee
2020 - 2021	**Karen Wang** Role: Chair Master of Architecture Study of thermal insulation strategies for Taiwanese housing units Committee: Rob Peña
2020 - 2021	**Chuou Zhang** Role: Chair Master of Architecture Design of mass-timber building components for reusability Committee: Kimo Griggs
2020	**Gayle Elam** Role: Committee member Master of Architecture "Acoustically Sensitive Large Assembly Spaces at School: An Elementary School Retrofit and Expansion" Chair: Rob Corser
2019 - 2020	**Vidhya Rajendran** Role: Chair Master of Science of Architecture - Design Computing "Helmholtz Resonators in Open Office Acoustics" Committee: Brian Johnson

IV.I Thesis Supervision at other Institutions

2017 - 2018 **Tomás Mena**
Master of Engineering in Integrated Design - Hochschule Ostwestfalen-Lippe
"Concept for intelligent facade configurator"
Committee: Hans Sachs, **Tomás Méndez Echenagucia**

2016 - 2017 **Jessica Menichelli**
Master of Architecture - Politecnico di Torino
"Optimization of the acoustic characteristics of diffusive surfaces : an objective evaluation method at the preliminary design phase"
Committee: Arianna Astolfi, Louena Shtrepi, and **Tomás Méndez Echenagucia**

2016 - 2017 **Giovanni Bouvet**
Master of Architecture - Politecnico di Torino
"Computational design : conchiglie acustiche parametriche per l'utilizzo contemporaneo dei teatri classici"
Committee: Arianna Astolfi, Elena Bo, and **Tomás Méndez Echenagucia**

2013 - 2014 **Denise Barbaroux**
Master of Architecture - Politecnico di Torino
"L'influenza delle balconate sulla qualità acustica delle sale da concerto"
Committee: Arianna Astolfi and **Tomás Méndez Echenagucia**

2012 - 2013 **Silvia Pastorino and Chiara Bertolutti**
Master of Architecture - Politecnico di Torino
"Studio acustico parametrico per la ricerca delle dimensioni ottimali di sale da concerto a "Shoebox" ed esagonali"
Committee: Arianna Astolfi, Louena Shtrepi, and **Tomás Méndez Echenagucia**

2012 - 2013 **Sabrina Canale**
Master of Architecture - Politecnico di Torino
"Lo studio dei riflettori acustici in una sala polivalente tramite simulazione ray-tracing"
Committee: Arianna Astolfi, Alessia Paola Griginis and **Tomás Méndez Echenagucia**

2012 - 2013 **Marco Palma and Maddalena Sarotto**
Master of Architecture - Politecnico di Torino
"Loudness : progettazione algoritmico-evolutiva di una conchiglia acustica per un palco rock"
Committee: Arianna Astolfi, Mario Sassone, Marco Carlo Masoero, Chiara Aghemo and **Tomás Méndez Echenagucia**

2012 **Fabio Cerniglia**
Master of Architecture - Politecnico di Torino
"Una casa itinerante"
Committee: Michele Bonino and **Tomás Méndez Echenagucia**

IV.J Applied Research Consortium supervision

This section contains work supervised under the Applied Research Consortium at the College of Built Environments - University of Washington. The work is often related to the thesis work of the same student, however it is always done on separate track with a more applied topic and a different committee. The work is done in collaboration with a Seattle based Architecture firm.

2021 - 2023 **Nathan Brown**
Structural design and fabrication of timber only joinery for structural reuse
Firm partner: Turner Construction - Josh Lohr, Sean Beatty
UW supervisor: **Tomás Méndez Echenagucia**

2020 - 2022 **Chuou Zhang**
Biogenic Carbon Accounting Method for Upstream Forest Factors: A Regional Approach
Firm partner: ZGF Architects - Jacob Dunn, Marty Brennan
UW supervisors: Indroneil Ganguli and **Tomás Méndez Echenagucia**

2019 - 2020 **Vidhya Rajendran**
"Heavenly Rooms: On the use of Helmholtz resonators for Open Office acoustics"
Firm partner: NBBJ Design - Ryan Mullenix
UW supervisor: **Tomás Méndez Echenagucia**

IV.K Research Assistant supervision

2025-2026 **Hasibul Hossain**
University of Washington
full time RA
POD / LCA coding

2024-2025 **Mohammad Tabatabai**
University of Washington
full time RA
POD / LCA coding

2021 **Nathan Brown**
University of Washington
Summer hourly RA
Design and vibro-acoustic analysis of mass timber floor systems

2020 **Vidhya Rajendran**
University of Washington
Summer hourly RA
Design, fabrication and on-site measurement of a resonant absorption panel in Cross Laminated Timber

2020	**Nathan Brown** University of Washington Summer hourly RA Creation of Design technology teaching resources website
2018 - 2019	**Theodora Ravanidou** ETH Zürich Full-time RA Flexible formwork system for the HiLo project
2017	**Pieter Bieghs** ETH Zürich Full-time RA Parametric study to improve the structural performance of a funicular floor system
2016 - 2017	**Alessandro Dell'Endice** ETH Zürich Full-time RA Parametric modeling of masonry cross-vaults for 3d printed structural models
2016 - 2017	**Ioannis Mirtsopoulos** ETH Zürich Full-time RA Design and fabrication of a cable-net and fabric formwork system in large prototype

IV.L Guest Lectures in other UW Courses

Spring 2026	**Arch 592 - Research Methods** "Research Methods" Invited by Prof. Brian McLaren and Prof. Mehlika Inanici
Fall 2024	**Arch 592 - Research Methods** "Research Methods" Invited by Prof. Mehlika Inanici
Fall 2023	**Arch 592 - Research Methods** "Acoustic Metamaterials" Invited by Prof. Mehlika Inanici
Spring 2023	**Arch 592 - Research Methods** "Computational design for sustainable construction" Invited by Profs. Ann Huppert and Tyler Sprague
Winter 2023	**Design Symposium** "Computational geometry for sustainable construction" Invited by Prof. Rob Peña

Fall 2022	**Arch 562** - **Contemporary Architectural Theory** "Mass Timber Vibro-Acoustics" Invited by Prof. Brian McLaren
Fall 2021	**Arch 362** - **Architecture and Theory** "Orders, Types and Computational Design" Invited by Profs. Alex Anderson and Brian McLaren
Winter 2021	**Arch 360** - **Introduction to Architectural Theory** "Patterns, Structures and Orders" Invited by Profs. Alex Anderson and Kathryn Rogers Merlino
Winter 2020	**ME 525** - **Applied acoustics** "Use of Helmholtz resonators for office acoustics" Invited by Prof. Peter Dahl
Winter 2020	**Arch 362** - **Introduction to Architectural Theory** "Patterns, Structures and Orders" Invited by Profs. Alex Anderson and Brian McLaren
Winter 2020	**Arch 524** - **Design Technology V** "Types, constraints and parametric models" Invited by Prof. Rob Peña
Winter 2020	**Arch 528** - **Digital design for fabrication and construction** "The COMPAS framework" Invited by Jack Hunter
Autumn 2019	**Arch 520** - **Design Technology I** "Compression only structures" Invited by Profs. Rob Peña and Kimo Griggs
Autumn 2019	**Arch 592** - **Research methods** "Computational geometry for sustainable construction" Invited by Profs. Kate Simonen and Ann Huppert
Autumn 2019	**Arch 562** - **Contemporary architectural theory** "Models" Invited by Profs. Alex Anderson and Brian McLaren

IV.M Studio and Thesis reviews

2021 **Innovative Mid-rise Timber** - final review
Weitzman School of Design - University of Pennsylvania (remote)
Instructor: Prof. Masoud Akbarzadeh

2021 **TOPOLOGY + timber** - final review
Taubman College of Architecture and Urban Planning - University of Michigan
Instructor: Prof. Tsz Yan Ng

2021 **Innovative Mid-rise Timber** - mid review
Weitzman School of Design - University of Pennsylvania (remote)
Instructor: Prof. Masoud Akbarzadeh

2020 **Digital Futures 2020 workshop** - final reviews
ETH Zürich - Department of Architecture (remote)
Instructors: Dr. Juney Lee, Dr. Tom van Mele and Prof. Philippe Block

2016 **Structural design II** - final review
ETH Zürich - Department of Architecture
Instructors: Prof. Philippe Block

2015 **Design Research Laboratory** - studio review
Architectural Association - London
Instructor: Shajay Bhooshan

2015 **Structural design II** - final review
ETH Zürich - Department of Architecture
Instructors: Prof. Philippe Block

IV.N Coding workshops

2021 **Python workflows for performance based design**
Hochschule Ostwestfalen-Lippe (remote)
Performance based methods for the early phase of design by means of the COMPAS framework

2018 **COMPAS_TNA workshop**
IASS 2018 workshops - MIT
Structural form-finding methods from compression only structures using Thrust Network Analysis

2018 **COMPAS_TNA workshop**
ETH Zürich - Department of Architecture
Masonry structures analysis using Thrust Network Analysis

2018 **COMPAS_FEA workshop**
ETH Zürich - Department of Architecture
Finite element analysis by means of the COMPAS framework

V Service

V.A Academic service positions at the University of Washington

2023 - ongoing **Executive committee**
Role: Committee member
Department of Architecture

2023 - ongoing **Design Technology committee**
Role: Chair
Department of Architecture

2023 - ongoing **UW Center for Digital Fabrication**
Role: Co-Director
Center Website
University of Washington

2022 - ongoing **Strategic planning committee**
Role: Committee member
Department of Architecture

2025 - 2026 **MSDT Program Evaluation Working Group**
Role: Chair
Department of Architecture

2025 - 2026 **Curriculum-Enrollment & Futures Working Group**
Role: Committee member
Department of Architecture

2020 - 2023 **Design Technology committee**
Role: Committee member
Department of Architecture

2021 - 2022 **Cohort hire committee**
Role: Committee member
Department of Architecture

2021 **FabLab director hiring committee**
Role: Committee member
College of Built Environments

2021 - 2022 **CBE Technology group**
Role: Committee member
College of Built Environments

2020 **MS Design computing admissions**
Role: Chair
Department of Architecture

2020 **Strategic planning review team**
Role: Committee member
College of Built environments

2019 - 2020 **Strategic planning Technology Task group**
Role: Committee member
College of Built environments

V.B Leadership positions in external organizations

2019 - ongoing **Open Research in Acoustical Science and Education**
Role: Board member
Non-profit for the support of acoustics education and research in the US
Founding board member

V.C Affiliations

- Acoustical Society of America (ASA)
- Association for Computer Aided Design in Architecture (ACADIA)
- International Association for Spatial and Shell Structures (IASS)